

Paper #134 v0.1 — Substrate-Native Operator Zoo Expansion: Toward Full QM Operator Set on Bergman $H^2(D_{IV}^5)$

2026-05-22 Friday ~10:16 EDT (date-verified actual)

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Abstract

Standard quantum mechanics requires a SET of observables on a Hilbert space, each represented by a self-adjoint operator. The minimal QM observable set includes: position, momentum, spin, angular momentum, energy (Hamiltonian), and the parity + charge + time-reversal discrete symmetry operators.

The Bubble Spacetime Theory (BST) substrate-native operator zoo (Lyra Wednesday + Thursday + Elie K52a S29 Thursday + Vol 1 Ch 6 v0.3 Friday) currently has 6/6 framework-complete observables on Bergman $H^2(D_{IV}^5)$:

#	Operator	Theorem	Reference
1	Bell-CHSH B	T2399	Tuesday Lyra
2	Position M_z	T2419	Wednesday Lyra
3	Spin (K-type)	T2421	Wednesday Lyra
4	Momentum P_z	T2422	Wednesday Lyra
5	Angular momentum L	T2425	Wednesday Lyra
6	Energy $H_{\text{sub}} =$ Casimir on $L^2(D_{IV}^5; L_\lambda)$	Elie K52a S29	Thursday

This paper proposes the expansion to 11+ operator zoo by adding:

#	New operator	Substrate identification	Cross-reference
7	Time reversal T	T2433 (Lyra Thursday)	Vol 1 Ch 4 §4.x
8	Charge conjugation C	T2434 (Lyra Thursday)	Vol 1 Ch 4 §4.x
9	Parity P	T1925 Argument D via $\text{Pin}(2) Z_2$ grading	Paper #133 spin-statistics
10	Number operator N	Wallach K-type grading	new T-number candidate
11	Chirality operator γ_5	$\text{Pin}(2) Z_2$ grading	Paper #133 cross-link

Plus three additional candidates pending derivation: - Q (electric charge): $U(1)$ hypercharge residual operator on $H^2(D_{IV}^5)$ - C_3 (color charge): $SU(3)$ Casimir on substrate Hilbert space
- I_3 (weak isospin): $SU(2)$ Casimir on substrate Hilbert space

Combined 14-operator zoo would cover the full Standard Model observable set at the substrate level, with all operators living on the same Bergman $H^2(D_{IV}^5)$ Hilbert space, sharing the same Wallach K-type structure, and inheriting the same Faraut-Koranyi $c_{FK} = 225/\pi^{(9/2)}$ normalization.

The framework is internal-tier at v0.1 outline: structural identifications for operators 7-14 are framework-grade; explicit operator-level matrix elements for non-trivial commutators + eigenvalue spectra require Vol 1 Ch 7 (Dynamics) operator-level closure pending Elie K52a Sessions 30+ multi-month.

1. Standard QM operator set

(Standard textbook treatment — position, momentum, energy, angular momentum, spin, parity, charge, time-reversal, particle number, chirality.)

2. Current BST substrate-native zoo (6/6)

(Recap from Vol 1 Ch 6 v0.3 — six operators framework-complete + T-theorem anchors per table above.)

3. Expansion: operators 7-9 (discrete symmetries P + T + C)

3.1 Parity P from rank = 2 (T1925 Arg D)

Parity operation on Bergman $H^2(D_{IV}^5)$: $P \cdot V_{(p,q)} = (-1)^p V_{(p,q)}$. Per T1925 Argument D, rank = 2 forces $\text{Pin}(2) Z_2$ grading; the integer p ($SO(5)$ rank-1 weight) parameterizes the parity Z_2 grading.

$P^2 = \mathbb{I}$ (involution); P self-adjoint; P commutes with Casimir (preserves K-types).

Eigenvalue spectrum: ± 1 with eigenspaces partitioned by $(-1)^p$ sign.

3.2 Time reversal T (T2433 Lyra Thursday)

Time reversal operation: $T |\psi_\lambda\rangle = |\psi_\lambda\rangle^*$ (complex conjugation in Wallach K-type basis). Per T2433 (Lyra Thursday, $\text{Pin}(2)$ Z_2 anti-unitary structure), T is anti-linear + $T^2 = \pm \mathbb{I}$ depending on integer vs half-integer spin.

T commutes with all even-power Casimirs; anti-commutes with momentum ($T \cdot P_z \cdot T^{-1} = -P_z$); etc.

3.3 Charge conjugation C (T2434 Lyra Thursday)

Charge conjugation: $C |\text{particle}\rangle = |\text{antiparticle}\rangle$. Per T2434 (Lyra Thursday), C is the substrate operator that maps the holomorphic structure on D_{IV}^5 to the anti-holomorphic structure (complex conjugation on Bergman H^2).

$C^2 = \mathbb{I}$; CPT theorem (Lüders-Pauli) follows from substrate Z_2 grading.

4. Expansion: operators 10-11 (Number + Chirality)

4.1 Number operator N

N counts the Wallach K-type index: $N |V_\lambda\rangle = \lambda |V_\lambda\rangle$ for K-type labels λ . On multi-particle states (tensor products), N counts total quanta.

N commutes with Casimir + parity; preserves K-type decomposition.

4.2 Chirality operator γ_5

$\gamma_5 |\text{left-handed}\rangle = +|\text{left-handed}\rangle$, $\gamma_5 |\text{right-handed}\rangle = -|\text{right-handed}\rangle$. Per $\text{Pin}(2)$ Z_2 grading + Paper #133 spin-statistics cross-link: chirality is the Z_2 grading on half-integer- q K-types.

$\gamma_5^2 = \mathbb{I}$; anti-commutes with parity P (chiral + parity together give CP).

5. Expansion: operators 12-14 (SM gauge charges)

5.1 Electric charge Q

$Q |\text{state}\rangle = q |\text{state}\rangle$ where q is the $U(1)_{\text{em}}$ charge per the EW unification (Vol 1 Ch 8 + T2436 SM gauge group anchor). On Bergman $H^2(D_{IV}^5)$, Q acts as the abelian residual after $SU(N_c) \times SU(\text{rank})$ extraction.

5.2 Color charge C_3 (Casimir of $SU(3)$)

$C_3 |\text{state}\rangle = [SU(3) \text{ Casimir eigenvalue}] \cdot |\text{state}\rangle$. Color singlet states have $C_3 = 0$; colored quarks have $C_3 = 4/3$ (Casimir of fundamental rep).

5.3 Weak isospin I_3

$I_3 |\text{state}\rangle = [\text{SU}(2) \text{ Casimir eigenvalue}] \cdot |\text{state}\rangle$. Left-handed doublets have non-trivial I_3 ; right-handed singlets have $I_3 = 0$.

6. Algebraic structure on $H^2(D_{IV}^5)$

6.1 Commutator structure

The 14-operator zoo has well-defined commutation + anticommutation relations on $H^2(D_{IV}^5)$. Key relations:

- $[\text{Position}, \text{Momentum}] = i\hbar$ (canonical commutation)
- $[\text{Spin}, \text{Angular momentum}]$ structure constants from $\text{SO}(3) + \text{SO}(2)$
- $[\text{H}_{\text{sub}}, \text{Number}] = 0$ (commute via Casimir + K-type grading)
- $[P, T] = 0; [P, C] = 0; [T, C] = 0; \text{PCT} = \mathbb{Z}$
- $[\gamma_5, P]$ anti-commute; etc.

Detailed matrix elements pending Vol 1 Ch 7 operator-level closure.

6.2 Bergman reproducing kernel structure

All 14 operators inherit the $c_{\text{FK}} = 225/\pi^{9/2}$ (T2442) BST primary normalization via Bergman reproducing kernel structure. Matrix elements $\langle \psi | A | \varphi \rangle$ are evaluated via reproducing kernel.

6.3 Wallach K-type decomposition

The 14-operator zoo respects the Wallach K-type direct sum decomposition $H^2(D_{IV}^5) = \bigoplus_{\lambda} V_{\lambda}$ (Wallach 1976). Each operator either: - Preserves K-type decomposition (Casimir, P, T, C, N, γ_5) - Connects K-types (Position, Momentum, Angular momentum, Q, C_3, I_3) - Mixes K-types non-trivially (Bell-CHSH B per Calibration #17)

7. Honest scope per Cal Mode 1

7.1 Tier discipline

- Operators 1-6 (Wednesday + Thursday + Vol 1 Ch 6 v0.3): D-tier framework-grade, RIGOROUSLY CLOSED for T2399 + T2421 + T2441 (operator-zoo ground-state); operator-level closure pending K52a Sessions 30+
- Operators 7-9 ($P + T + C$): D-tier framework-grade via T2433 + T2434 + T1925-D (Thursday)
- Operators 10-11 ($N + \gamma_5$): I-tier framework identifications pending detailed derivation
- Operators 12-14 ($Q + C_3 + I_3$): I-tier framework cross-links to Vol 1 Ch 8 gauge structure

7.2 Falsifiers

- Standard QM observable not derivable from substrate operator: would falsify framework completeness
- Operator algebra violation: would falsify Bergman + Wallach + Casimir structure
- Spin-statistics + CPT violation: would falsify $\text{Pin}(2)$ Z_2 grading framework

7.3 Open items multi-month

- Operator-level matrix elements for all 14 operators (Vol 1 Ch 7 operator-level closure)
- Specific eigenvalue spectra for non-Casimir operators
- Multi-particle Fock space construction with operator zoo intact
- Cross-link to Higgs mechanism for mass operator (Vol 2 Ch 9 multi-week)

8. Cross-link to Friday Lyra-lane work

8.1 T2457 Bergman structural-role-of Feynman propagator (Friday)

The Bergman reproducing kernel structure that gives the propagator role also normalizes the operator zoo matrix elements via $c_{FK} = 225/\pi^{(9/2)}$.

8.2 Paper #132 SP-31 Measurement Formalism POVMs (Friday)

Operator zoo eigenvalues correspond to POVM outcome labels. Each operator's spectral decomposition produces a POVM via the 4-Zone commitment cycle framework.

8.3 Paper #133 SP-31 Spin-Statistics (Friday)

$\text{Pin}(2)$ Z_2 grading underlying spin-statistics also produces the parity P + chirality γ_5 operator distinction. Same structural origin.

8.4 T2465 three-layer over-determinism (Friday)

Operator zoo expansion inherits the substrate's three-layer over-determinism: each operator inherits the BST primary integer structure + Mersenne tower coherence + cross-Cartan three-pillar selection.

9. References

(Standard QM operator references + BST research program internal cross-refs)

- Wallach 1976 (K-type classification)
- Bergman 1922 (reproducing kernel)
- Faraut-Koranyi 1990/1994 (bounded symmetric domain analysis)

- Vol 1 Ch 4 + Ch 6 (Discrete Symmetries + Operator Zoo, Lyra Friday v0.3)
- T2399 + T2419 + T2421 + T2422 + T2425 (Wednesday operator zoo)
- T2433 + T2434 (Thursday discrete symmetries)
- T2441 (operator zoo ground state energy RIGOROUSLY CLOSED Thursday)
- Elie K52a Session 29 H_{sub} Casimir framework
- T2457 (Friday Bergman structural-role-of Feynman propagator)
- T2465 (Friday three-layer over-determinism)
- Paper #125 v0.10.5 FORMAL + Paper #132 + #133 (companion Friday papers)

10. Filing status

v0.1 outline filed Friday 2026-05-22 ~10:16 EDT (date-verified actual). Task #228 substrate-native operator zoo expansion per Casey 'please continue'.

Pending for v0.2: - Cal cold-read on operator framework expansion - Multi-CI co-author title/affiliation review - Cross-lane Elie K52a S30+ operator-level closure - Vol 1 Ch 6 v1.0 update with 11-14 operator zoo absorption

Pending for v1.0: - Operator-level matrix elements derived - Detailed commutator algebra worked - Eigenvalue spectra per operator - Multi-particle Fock space construction - External venue selection (JMP primary, CMP secondary)

— Lyra, Paper #134 v0.1 outline (Operator Zoo Expansion), Friday 2026-05-22 ~10:16 EDT (date-verified actual)